

CPI INFLATION ANALYSIS

2013 – 2023

An Excel & Power Query–based analysis of India's Consumer Price Index, covering basket contribution, inflation trends, food-price drivers, COVID-19 impact, and oil-price correlation.

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TOOLS

Excel 2024 · Power Query · Pivot Tables · Pivot Charts · Correlation Analysis

VERSION

2.0 (Enhanced)

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Executive Summary

This project analyzes India's Consumer Price Index (CPI) data from 2013–2023 to uncover inflation trends, category-level contributions, the impact of COVID-19, key drivers of food inflation, and the relationship between CPI movements and imported oil prices. The full workflow — from raw data cleanup to business-question analysis — was built in Excel using Power Query, Pivot Tables, Pivot Charts, and correlation analysis.

Business Questions Solved

- How much does each broader category contribute to the overall CPI basket?
- How has year-on-year inflation trended since 2017, and which year peaked?
- What drove food inflation over the 12 months ending May 2023?
- How did the onset of COVID-19 affect inflation across key categories?
- Which category's inflation correlates most strongly with imported oil prices?

Data Cleaning & Transformation

Before analysis, the raw All-India CPI index was cleaned and reshaped in Power Query so it could support Pivot Table and trend analysis:

- ✓ Replaced "NA" text values with proper nulls
- ✓ Fixed missing Housing values using Fill Up
- ✓ Filled missing CPI values using Fill Down
- ✓ Corrected month-name spelling errors
- ✓ Trimmed text inconsistencies across category labels
- ✓ Created a Month Number field for chronological sorting
- ✓ Unpivoted the wide-format table into an analysis-ready long format

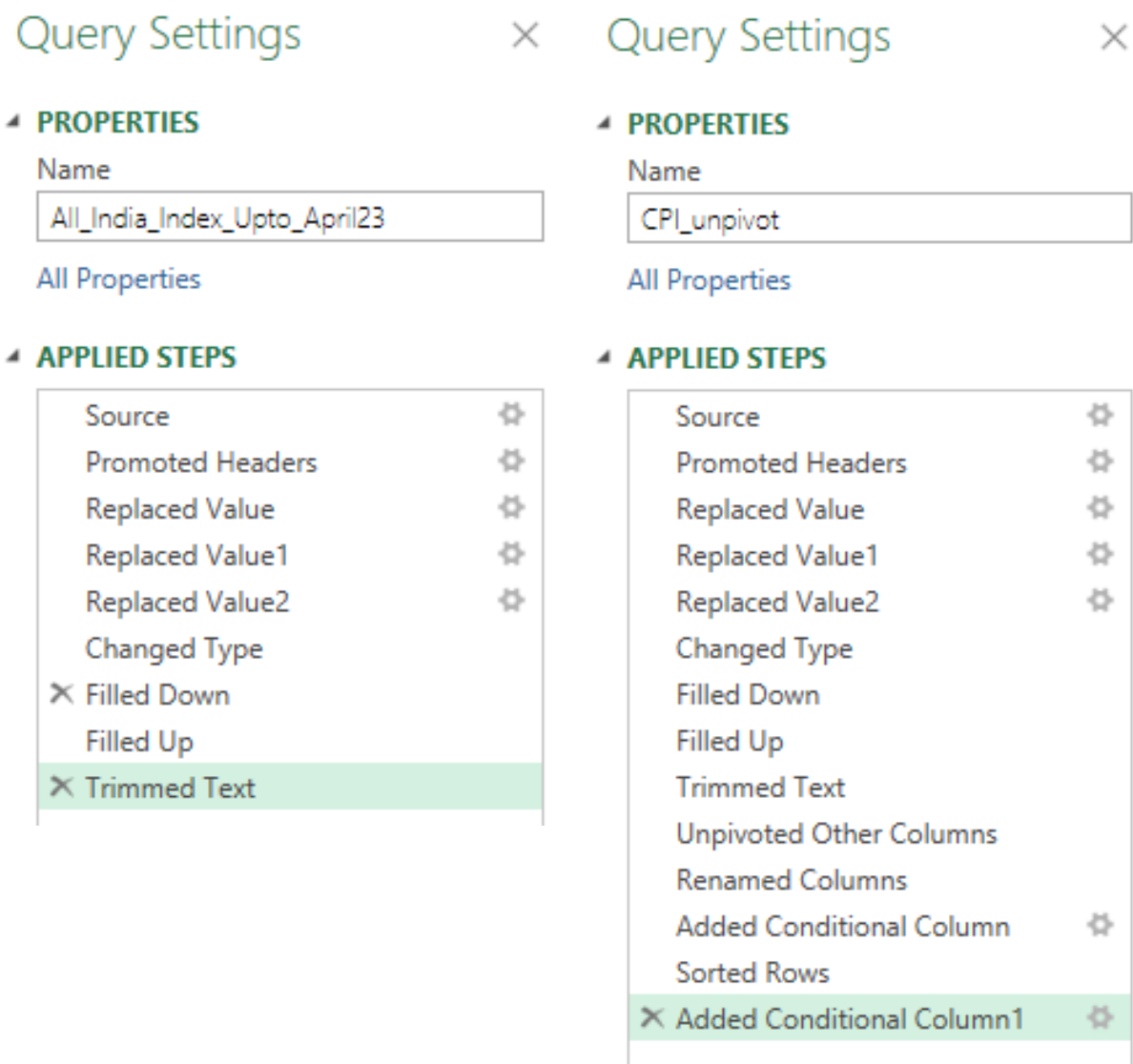


Fig. 1 — Cleaning steps applied to the All-India index query

Fig. 2 — Steps applied to the unpivoted CPI query

Analytical Data Model

The dataset was transformed from a wide format into a normalized, long-format analytical structure — one row per Sector × Year × Month × Category — suitable for Pivot Table and trend analysis. A sample of the resulting table is shown below.

Sector	Year	Month	Categories	Indexes	Broader Categories	Month Number	Date	Month_M
Rural	2013	January	Vegetables	101.9	Food and Beverages	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Vegetables	102.2	Food and Beverages	1	01-01-2013	Jan-2013
Urban	2013	January	Vegetables	102.9	Food and Beverages	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Transport and communication	103.2	Transport and communication	1	01-01-2013	Jan-2013
Rural	2013	January	Transport and communication	103.3	Transport and communication	1	01-01-2013	Jan-2013
Urban	2013	January	Transport and communication	103.2	Transport and communication	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Sugar and Confectionery	106.2	Food and Beverages	1	01-01-2013	Jan-2013
Rural	2013	January	Sugar and Confectionery	106.8	Food and Beverages	1	01-01-2013	Jan-2013
Urban	2013	January	Sugar and Confectionery	105.1	Food and Beverages	1	01-01-2013	Jan-2013
Rural	2013	January	Spices	103.1	Food and Beverages	1	01-01-2013	Jan-2013
Urban	2013	January	Spices	101.8	Food and Beverages	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Spices	102.7	Food and Beverages	1	01-01-2013	Jan-2013
Urban	2013	January	Recreation and amusement	102.9	Miscellaneous	1	01-01-2013	Jan-2013
Rural	2013	January	Recreation and amusement	103.4	Miscellaneous	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Recreation and amusement	103.1	Miscellaneous	1	01-01-2013	Jan-2013
Urban	2013	January	Pulses and products	105.8	Food and Beverages	1	01-01-2013	Jan-2013
Rural	2013	January	Pulses and products	106.1	Food and Beverages	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Pulses and products	106	Food and Beverages	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Prepared meals, snacks, sweets etc.	107.3	Food and Beverages	1	01-01-2013	Jan-2013
Urban	2013	January	Prepared meals, snacks, sweets etc.	107.9	Food and Beverages	1	01-01-2013	Jan-2013
Rural	2013	January	Prepared meals, snacks, sweets etc.	106.7	Food and Beverages	1	01-01-2013	Jan-2013
Urban	2013	January	Personal care and effects	104.3	Personal care	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Personal care and effects	104.5	Personal care	1	01-01-2013	Jan-2013
Rural	2013	January	Personal care and effects	104.7	Personal care	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Pan, tobacco and intoxicants	105.1	Pan, tobacco and intoxicants	1	01-01-2013	Jan-2013
Rural	2013	January	Pan, tobacco and intoxicants	105.1	Pan, tobacco and intoxicants	1	01-01-2013	Jan-2013
Urban	2013	January	Pan, tobacco and intoxicants	105.2	Pan, tobacco and intoxicants	1	01-01-2013	Jan-2013
Urban	2013	January	Oils and fats	103.4	Food and Beverages	1	01-01-2013	Jan-2013
Rural	2013	January	Oils and fats	106.1	Food and Beverages	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Oils and fats	105.1	Food and Beverages	1	01-01-2013	Jan-2013
Rural+Urban	2013	January	Non-alcoholic beverages	104.9	Food and Beverages	1	01-01-2013	Jan-2013

Fig. 3 — Sample of the cleaned, unpivoted CPI dataset (Jan 2013 shown)

1 Category Contribution Analysis

Food & Beverages contribute approximately 48% of the CPI basket, making them the most influential category.

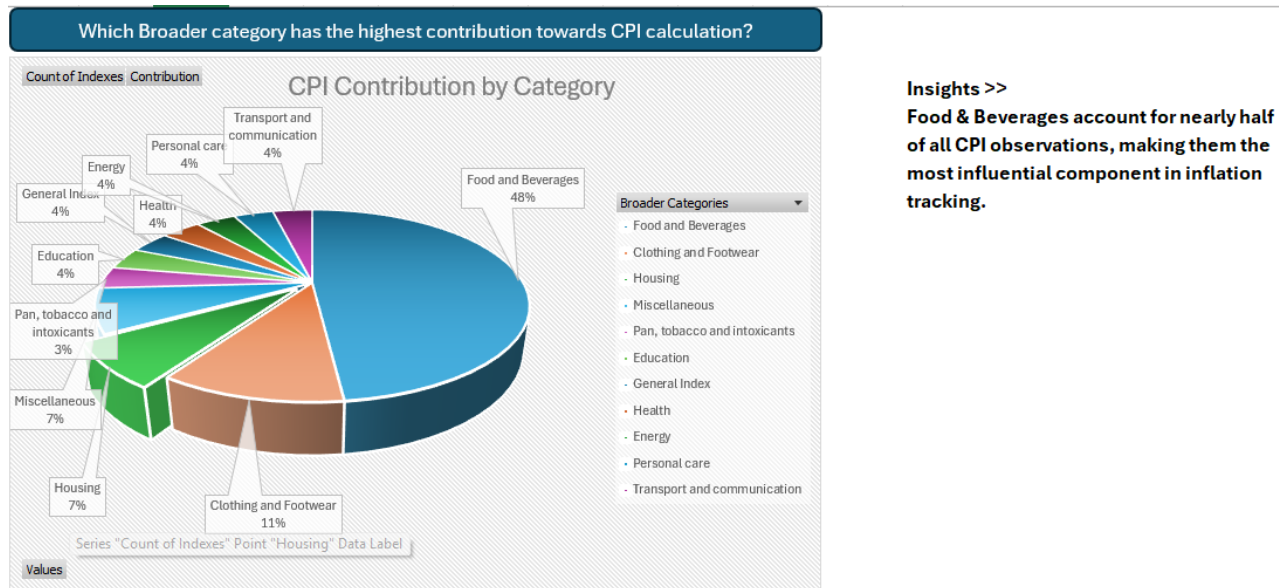
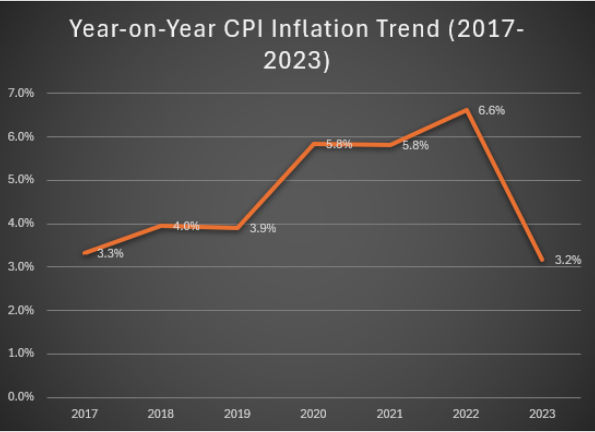


Fig. 4 — Broader-category contribution to the CPI basket

2 Year-on-Year Inflation Trend

2022 recorded the highest inflation rate due to supply-chain disruptions, rising oil prices, and post-pandemic recovery.

A trend of Y-o-Y increase in CPI (rural + urban) inflation starting 2017 for the entire basket of products combined.



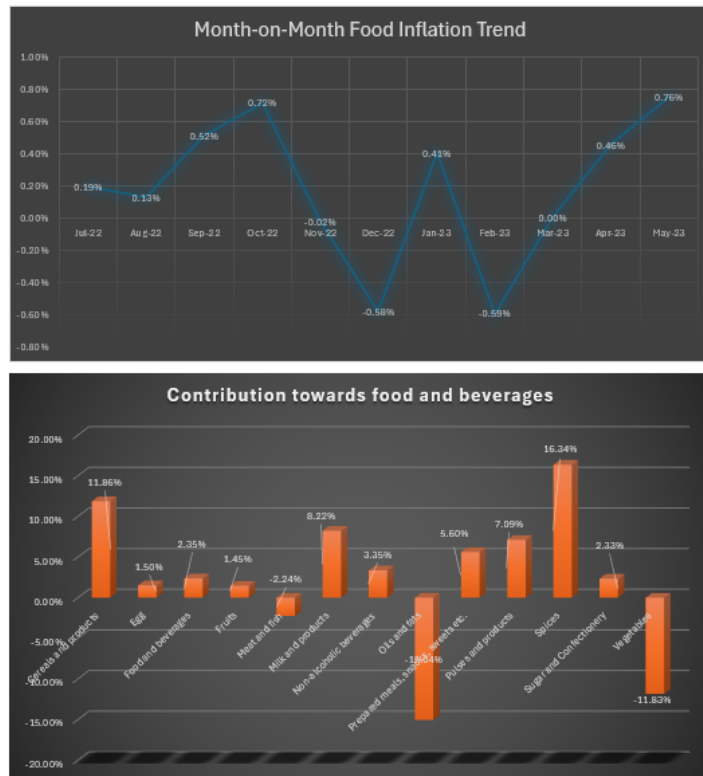
Insights>>
Year 2022 has the highest inflation rate at 6.6%
There was major issues going on in year 2022 where Russia and ukraine war was declared which impacted the rise in the crude oil prices, edible oil spikes.
The highest inflation period coincided with:
Global supply chain disruptions
Rising crude oil prices
Post-pandemic demand recovery
Food price inflation
These factors increased transportation and commodity costs, leading to higher CPI inflation.

Fig. 5 — Year-on-year CPI inflation, 2017–2023

3 Food Inflation Analysis

May 2023 recorded the highest food inflation of the trailing 12 months, while spices were the largest contributor.

With India's retail inflation reaching a 3-month high of 5.55% in November 2023, largely due to a sharp rise in food prices. Analyze the following for 12 months ending May'23



Insights>>

Highest food inflation in the month of May-23 at 0.76%

Lowest food inflation was in the month of Feb-23 at -0.59%

Absolute change - Food index increased by 3.46 points.

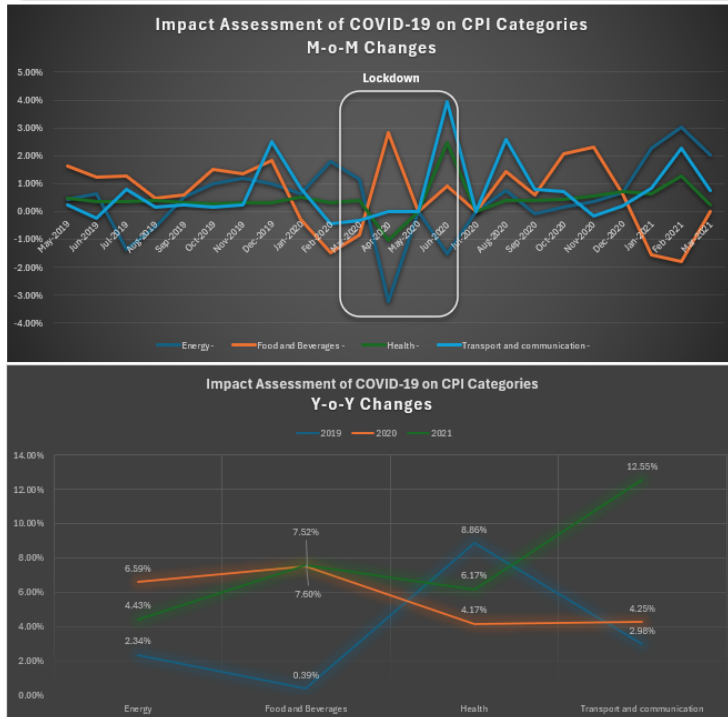
Spices are the highest contributor within food and beverages category at 16.44%

Fig. 6 — Month-on-month food inflation and category contribution

4 COVID-19 Impact Assessment

Healthcare, food, and transport categories experienced the most significant swings following the onset of COVID-19.

Investigate how the onset and progression of the COVID-19 pandemic affected inflation rates in India. Analyze the Impact of key pandemic milestone (first lockdown) on the CPI inflation %, specially focus on categories like healthcare, food, and essential services.



Insights >>

Food spiked in Apr 2020 (2.8%). Health peaked in Jun 2020 (2.4%)
Energy and Transport showed stability during lockdown
Food showed maximum volatility due to supply chain disruptions

Insights >>

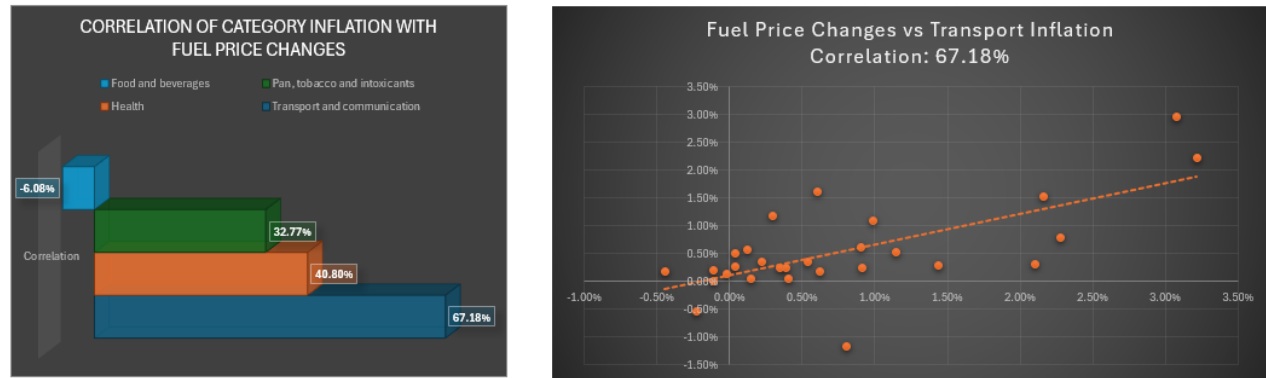
Transport showed biggest surge (3% → 13%) - V-shaped recovery
Health rose steadily (4% → 9%) - pandemic impact
Food remained elevated (7.9%) throughout

Fig. 7 — CPI category behaviour before, during, and after the first lockdown

5 Oil Price Correlation Analysis

Transport & Communication showed the strongest positive correlation with fuel price movements.

Identify the category whose inflation prices strongly change with fluctuations in imported oil price.



Insights >>
Transport and communication has the strongest correlation (67.18%) with fuel price changes.

Rank	Category	Correlation	Strength
1	Transport and communication	67.18%	Strong Positive
2	Health	40.80%	Moderate Positive
3	Pan, tobacco and intoxicants	32.77%	Weak Positive
4	Food and beverages	-6.08%	Negligible Negative

Fig. 8 — Correlation of category inflation with fuel price changes

Key Findings

- Food & Beverages dominate CPI basket contribution, at roughly 48% of the index.
- Inflation peaked in 2022 (6.6%) driven by global supply-chain and energy shocks.
- Spices were the largest single contributor to food inflation, at 16.44%.
- COVID-19 significantly disrupted the Health and Transport categories in particular.
- Transport & Communication showed the strongest relationship with fuel prices (67.18% correlation).

Skills Demonstrated

Excel

Power Query

Data Cleaning

Data Transformation

Pivot Tables

Trend Analysis

Correlation Analysis

Business Insight Generation